October 2026

Authorized for Public Release

Large Science Models:

Foundation Models for Generalizable Insights Into Complex Systems with Psycho-social Application

DARPA-EA-25-02-05-MAGICS-PA-025
HR0011-26-3-E016

Zero Knowledge Discovery

PI: Ishanu Chattopadhyay, PhD

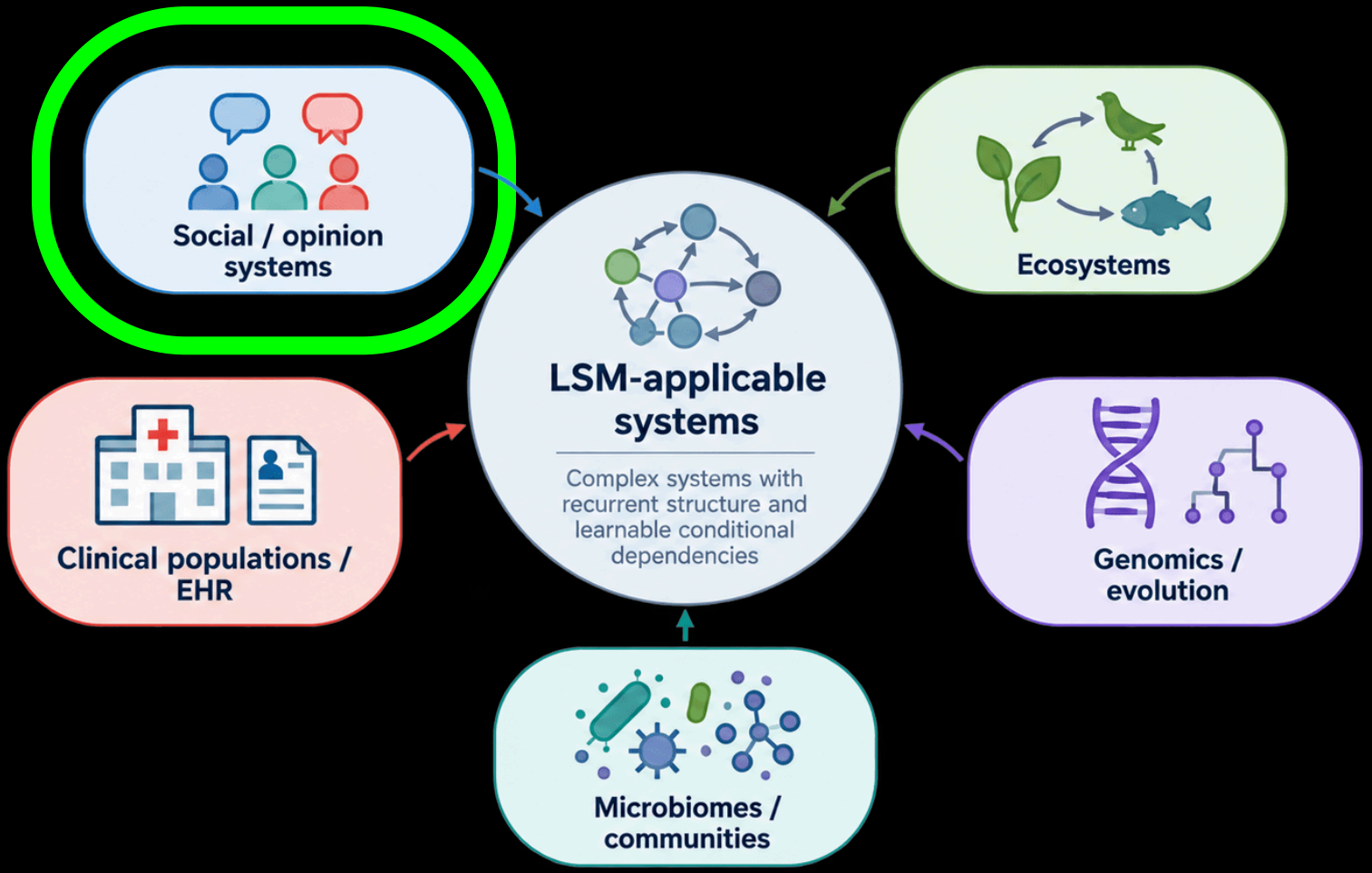
Assistant Professor of Biomedical Informatics & Computer Science

University of Kentucky

DARPA



LSMs for complex systems arising in non-sequential multivariate measurements



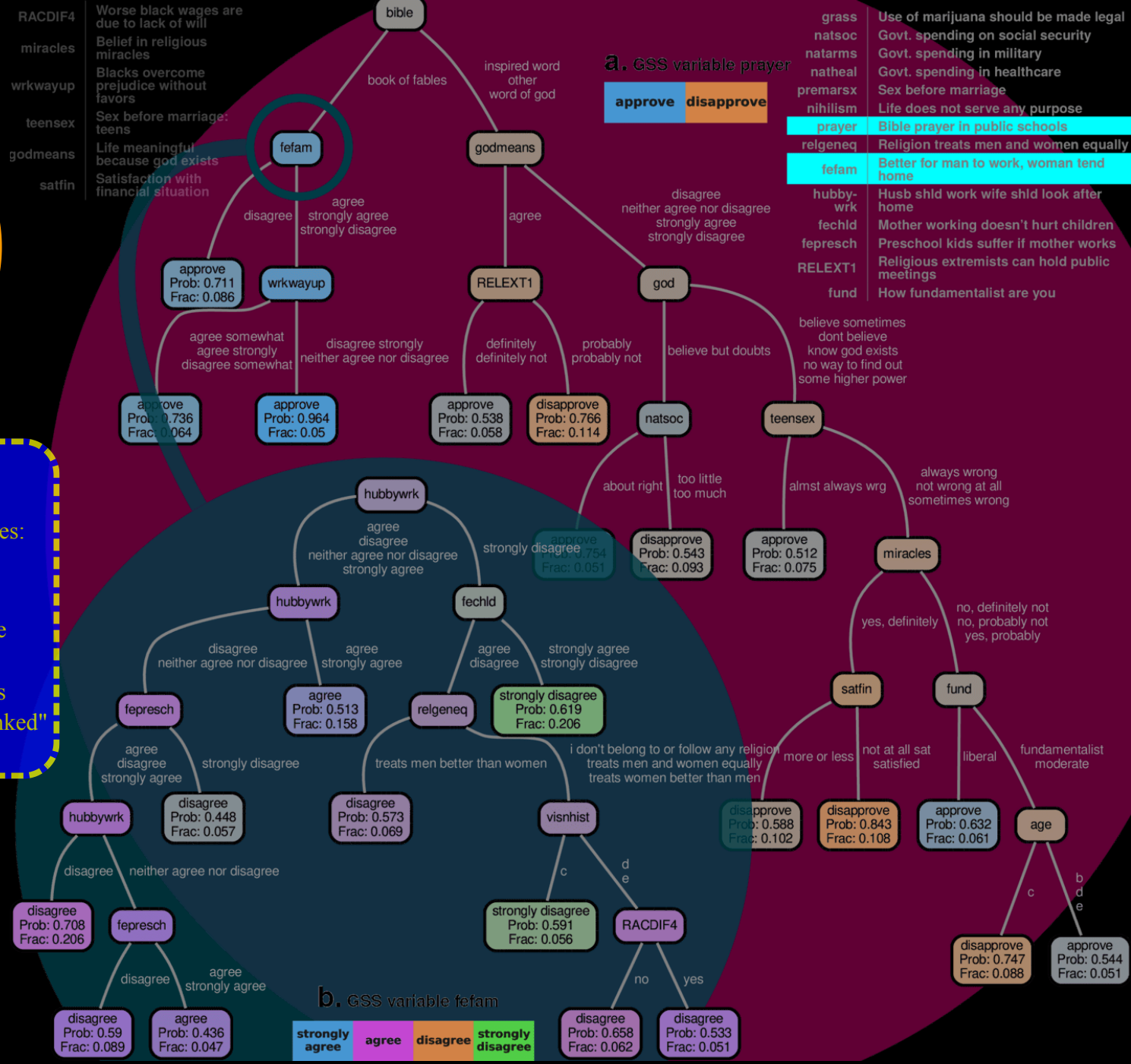
LSM Forest

GSS 2018 dataset

Recursive LSM forest: hyperlinked nodes capturing emergent macro-structures

- Set of ***conditional inference trees*** (CIT)
  - Strict statistical guarantees: quantifies inference uncertainty
- Each tree models exactly one variable as a function of potentially all other variables
- Non-leaf nodes are "hyperlinked" to other trees

✓ 1. How will proposer form and maintain a computationally tractable LSM tree structure given, as proposed, hundreds to thousands of observable variables?



Large Science Models

Computationally tractable LSM tree structure given, as proposed, hundreds to thousands of observable variables.

https://paraknowledge.ai/data/trees2018/obey.svg

GSS 2018 dataset

- Each predictor is inferred independently
- Can scale up to thousands of variables in Python implementation
- Further scale-up $10^6 - 10^8$ needs C/C++ implementation

Full Example of Hyperlinked Trees

https://34.66.189.202/data/trees2018/

The Empirical Foundation: Global Opinion Geometry

4.05M+
Participants

252 surveys "digitized"

1972–2025
Coverage

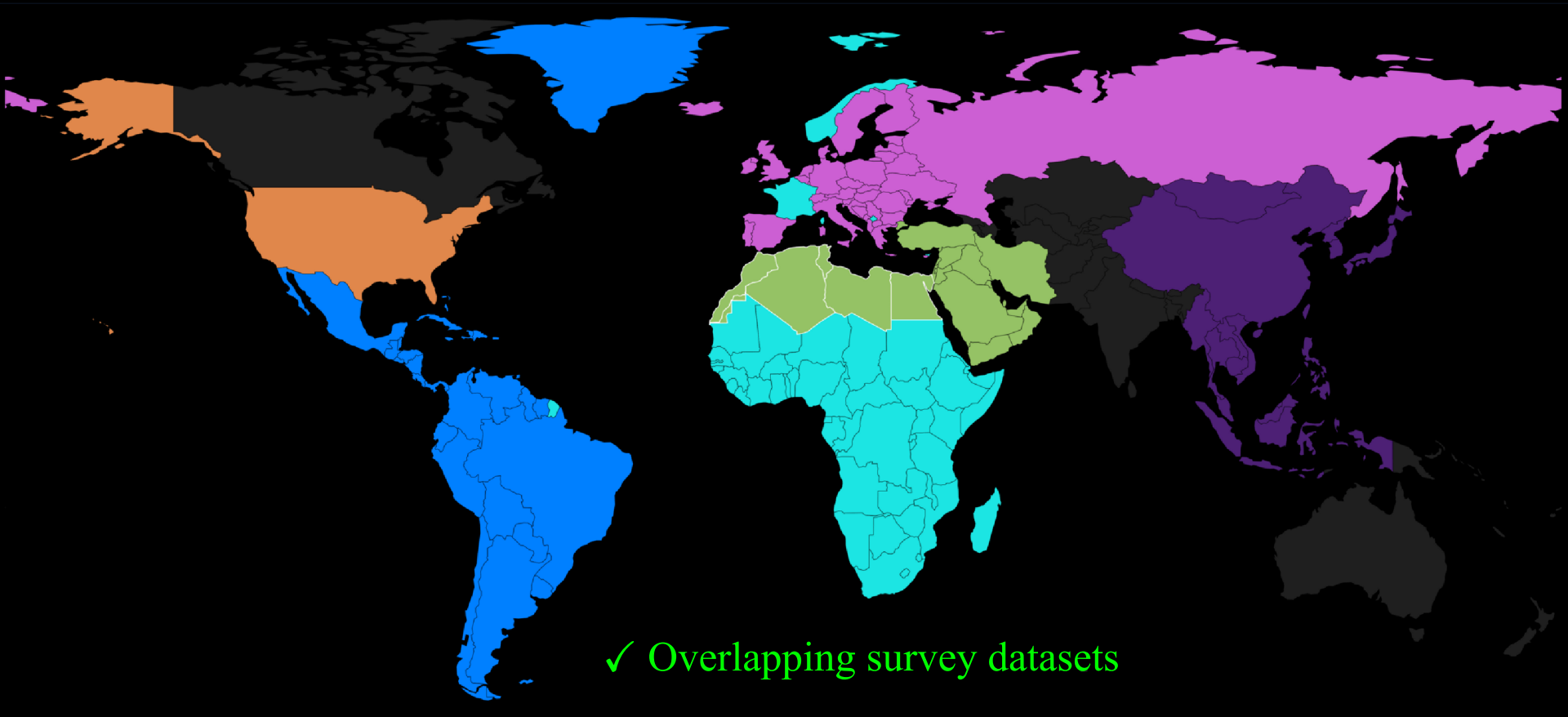


- GSS
- Eurobarometer
- Afrobarometer
- WVS



Insight: These combined
empirical geometry required
"Digital Twins" of global po
mapped by time, geography,

Goal: Global Coverage



Large Science Models: Mathematical Framework

Observables: $X = \{x^1, \ldots, x^N\}, \quad x^i \in \Sigma^i$ (finite alphabet)

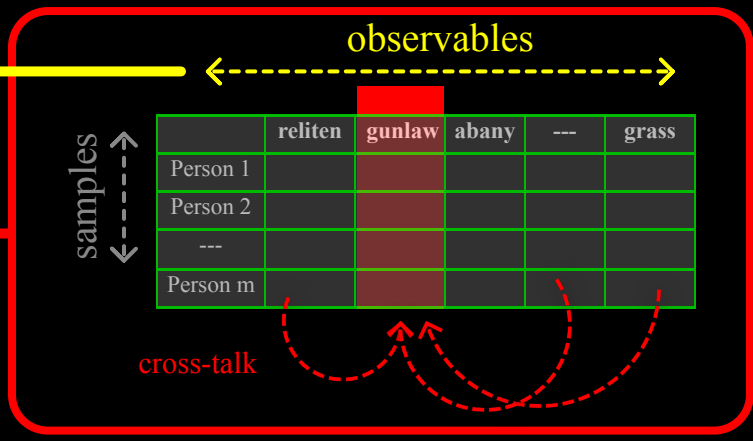
Notation: $x^{-i} = \{x^j : j \neq i\}$

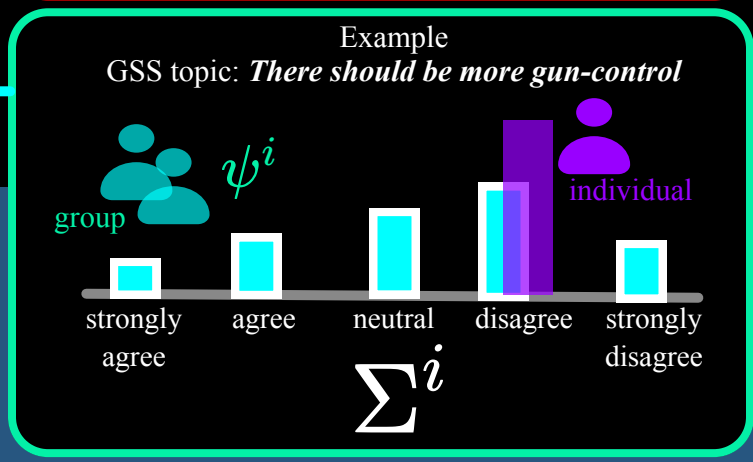
Crosstalk: $\forall i \; P(x^i) = f_i(x^{-i})$

System state: $\psi = \bigotimes_{i=1}^{N} \psi^i, \quad \psi^i \in \mathscr{D}(\Sigma^i) \cup \varnothing$ (missing observation)

Notation: $\psi^{-i} = \bigotimes_{j \neq i} \psi^j$

ψ^i: Distributions over alphabet Σ^i





Individual Predictor (CIT)

$\phi = \bigotimes_{i=1}^{N} \phi^i, \quad \phi^i(\psi^{-i}) \in \mathscr{D}(\Sigma^i)$

estimate is always a non-empty non-degenerate distribution

$\phi^i(\psi^{-i}) \sim \tilde{\psi}^i$

ϕ estimates ψ

$\phi(\psi) || \psi$

Tension between predicted and observed distribution drives change

Digital Twin

Examples: GSS, ANES, WVS, ESS, Eurobarometer, Afrobarometer, Asian Barometer etc

Large Science Models: Properties

LSM-Distance Metric*

$$\theta(\psi,\psi') \triangleq \frac{1}{N}\sum_{i=1}^{N}\sqrt{D_{JS}\left(\phi^i(\psi^{-i})||\phi^i(\psi'^{-i})\right)}$$

where $D_{JS}(P||Q)$ is the Jensen-Shannon divergence.

Induced Riemannian metric tensor

$$g_{ij}(\psi) = \frac{1}{2}\frac{\partial^2}{\partial\psi^i\,\partial\psi^j}\theta^2(\psi,\psi')\Big|_{\psi'=\psi}$$

This is a standard Physics construct, quantifying curvature of the underlying latent geometry

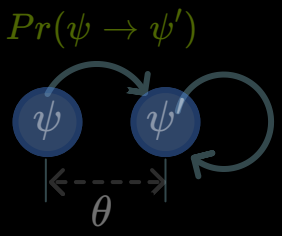
Large Deviation Bound*
(Sanov's Theorem, Pinkser's Inequality)

$$\left|\ln\frac{\Pr(\psi\to\psi')}{\Pr(\psi'\to\psi')}\right| \le \beta\,\theta(\psi,\psi')$$

$\Pr(\psi'\to\psi')$: persistence probability

β: const. scaling as N^2

This bound connects ``closeness" of samples to the odds of perturbing from one to the other, *bridging geometry to dynamics*



Ergodic Projection
"spatial average": average of all plausible worldviews or states

$$\psi_\star \triangleq \bigotimes_{i=1}^{N}\phi^i\left(\prod_{1}^{N-1}\varnothing\right)$$

Easily computable in LSM framework!

Start with opinion vector with all entries missing → Apply ϕ^i

Ergodic dispersion

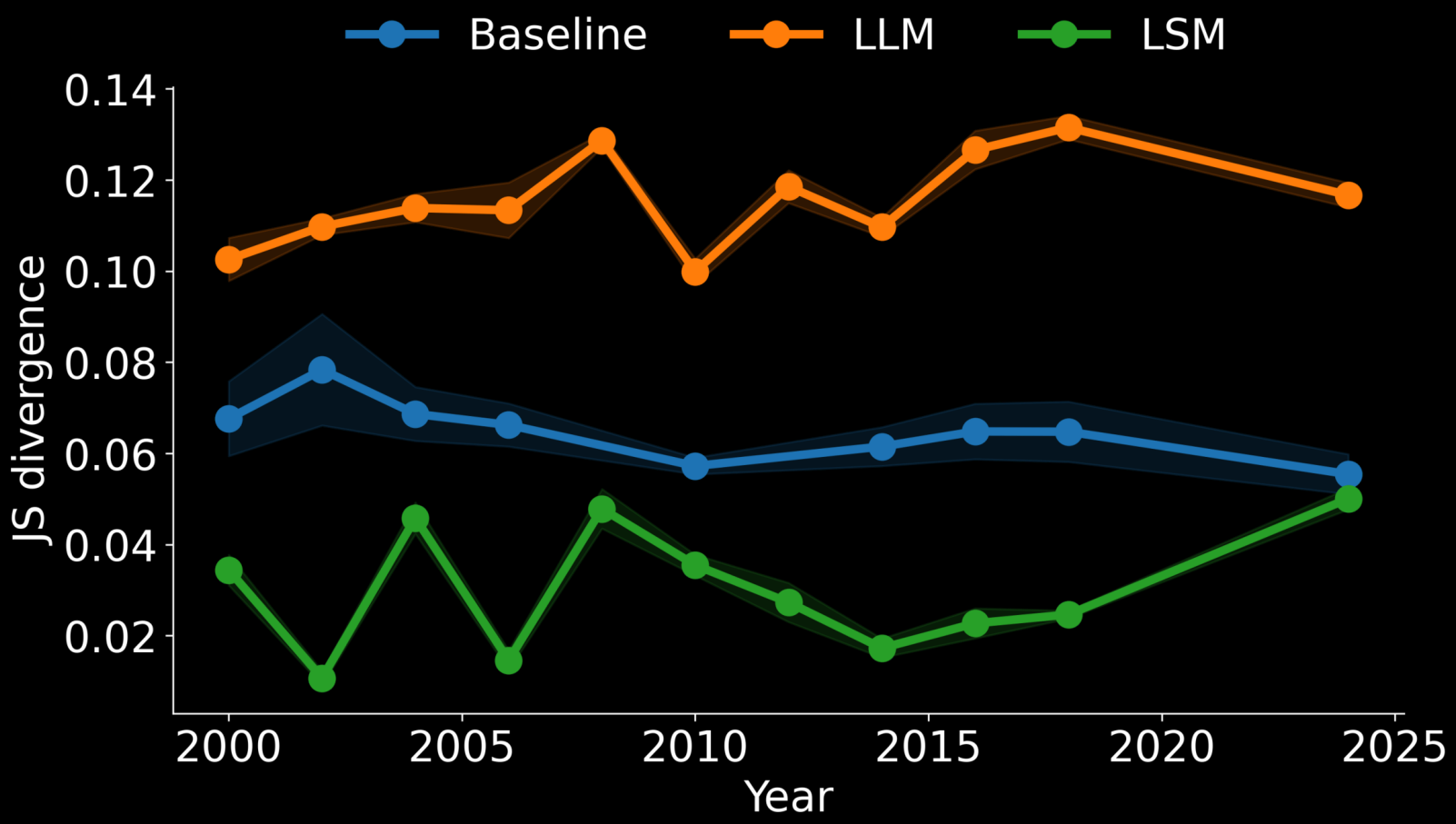
$$\Psi_\star = \theta(\psi,\psi_\star)$$

Random variable quantifying dispersion around the spatial average of worlviews

Central to Model Drift Quantification

* Sizemore, Nicholas, Kaitlyn Oliphant, Ruolin Zheng, Camilia R. Martin, Erika C. Claud, and Ishanu Chattopadhyay. "A digital twin of the infant microbiome to predict neurodevelopmental deficits." Science Advances 10, no. 15 (2024): eadj0400. https://www.science.org/doi/full/10.1126/sciadv.adj0400

Old Validation Strategy: Compare with LLMs

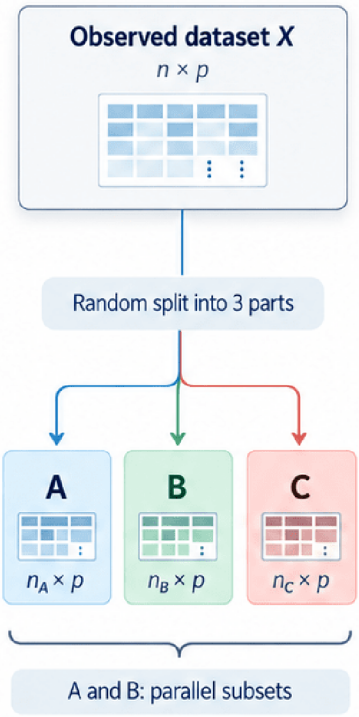


Validation?

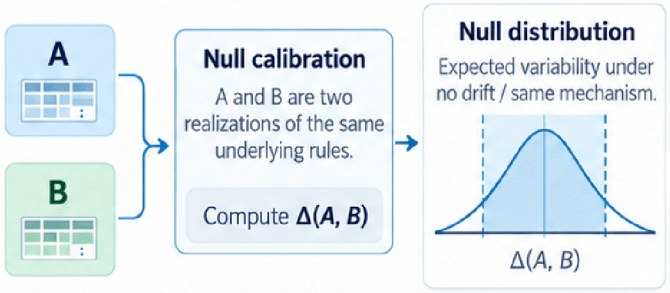
Validation of Data Drift and Synthetic Fidelity

Calibrating a null for expected variability, then locating synthetic and broken-structure controls relative to that null.

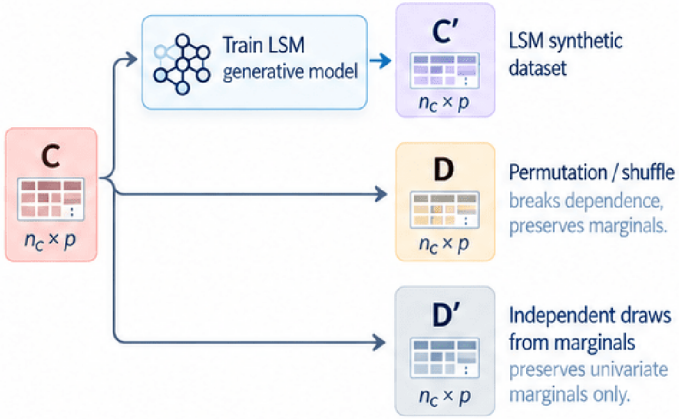
1 DATA SPLIT



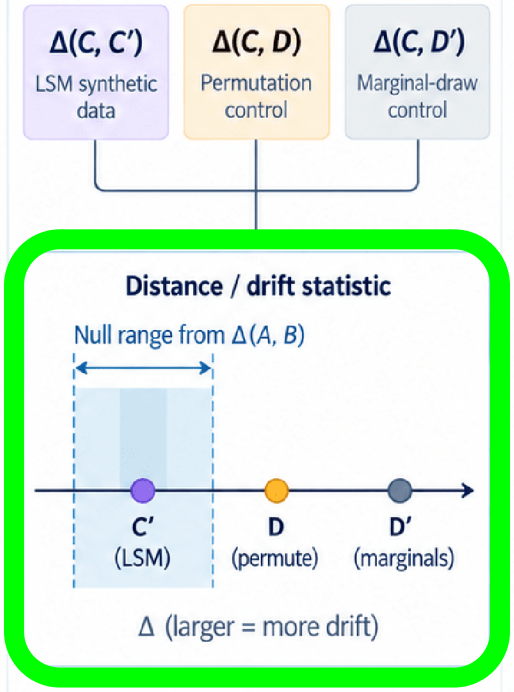
2 NULL CALIBRATION



3 MODEL + NEGATIVE CONTROLS



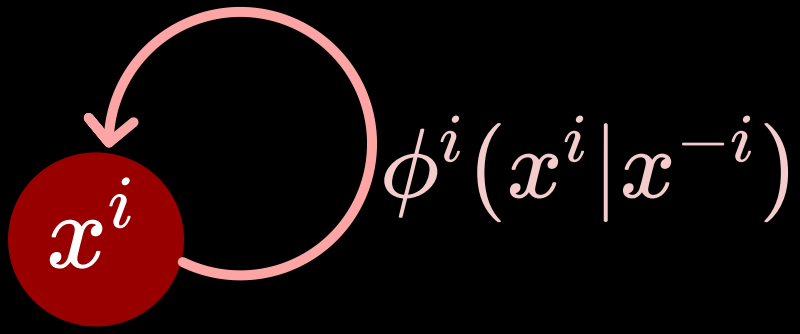
4 EVALUATE AGAINST THE NULL



Good synthetic data should lie near the null; broken-structure controls should lie outside it.

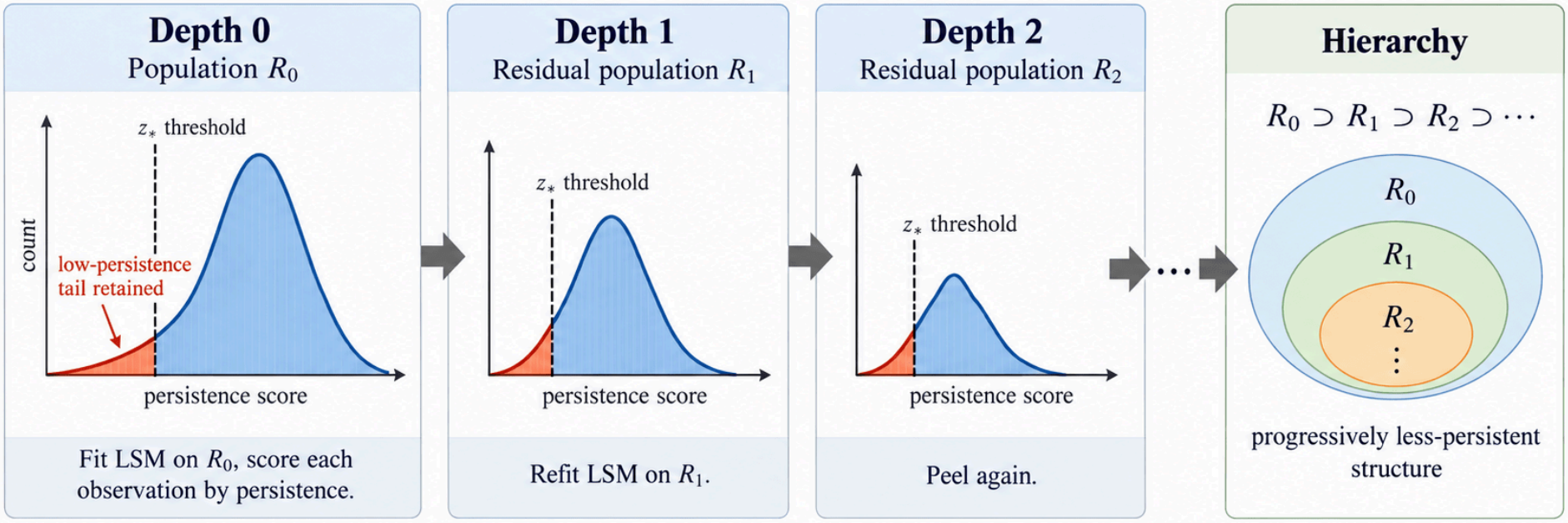
Sample Persistence:

$$\omega(x) = \frac{1}{n} \sum_{i=1}^{n} \log_2 \phi^i(x^i \mid x^{-i})$$



How well does sample x fit the model, or doesnot tend to perturb?

Multi-scale LSM vi Progressive Persistence Peeling



$$R_{d+1} = \{x \in R_d : z_d(x) > z_*\}$$

larger $z_d(x)$ = lower persistence

At each depth: fit an LSM → score persistence → retain the low-persistence tail → refit.

Fit an LSM to the current population:

$$M_0 = \mathrm{LSM}(R_0)$$

↓

Score each observation by its conditional persistence:

$$\omega_d(x) = \frac{1}{n}\sum_{i=1}^{n} \log_2 \phi_i^{(d)}(x_i \mid x_{-i})$$

↓

Peel the low-persistence tail:

$$R_{d+1} = \left\{ x \in R_d : \frac{\omega_d(x) - \mu_d}{\sigma_d} < -z_* \right\}$$

↓

Refit LSM

$$M_{d+1} = \mathrm{LSM}(R_{d+1})$$

Repeatedly isolate progressively less-persistent observations and relearn LSM

$$R_0 \supset R_1 \supset R_2 \supset \cdots$$

Fingerprinting LSMs

MAP error → $$\upsilon_{M_d}(x, i) = \frac{\phi_i^{(d)}(x_i \mid x_{-i})}{\max_a \phi_i^{(d)}(a \mid x_{-i})}$$

How likely was the actual outcome?

Conditional displacement:

$$\Delta C(d) = \mathbb{E}_x \left[\frac{1}{n} \sum_i |\upsilon_{M_d}(x, i) - \upsilon_{M_{d+1}}(x, i)| \right]$$

How many bits needed to describe the distributions?

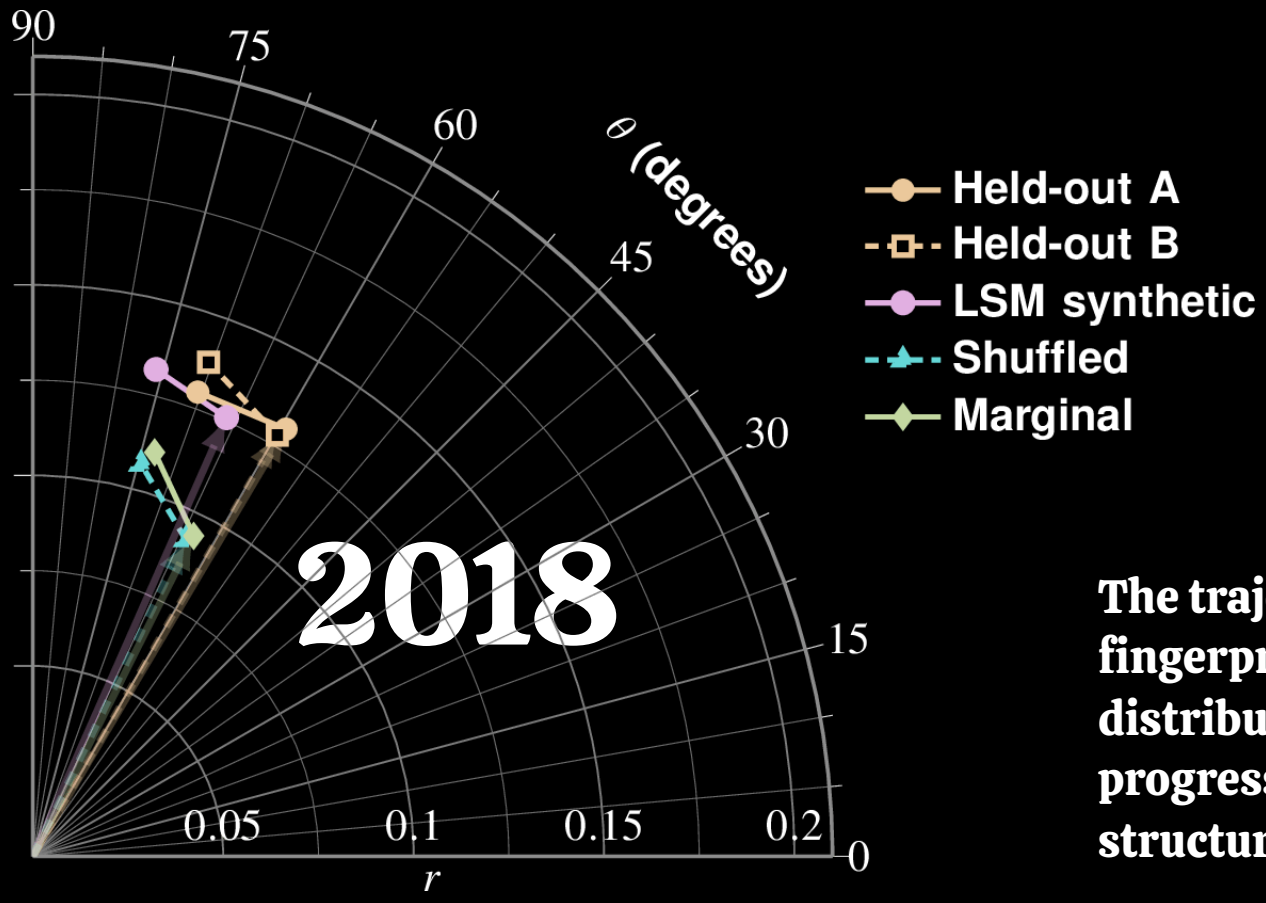
Description length → $$\ell_d(x) = -\frac{1}{n} \sum_i \log_2 \phi_i^{(d)}(x_i \mid x_{-i})$$

Information displacement:

$$\Delta K^{\text{pseudo}}(d) = \mathbb{E}_{R_{d+1}}[\ell_d] - \mathbb{E}_{R_d}[\ell_d]$$

Fingerprinting LSMs

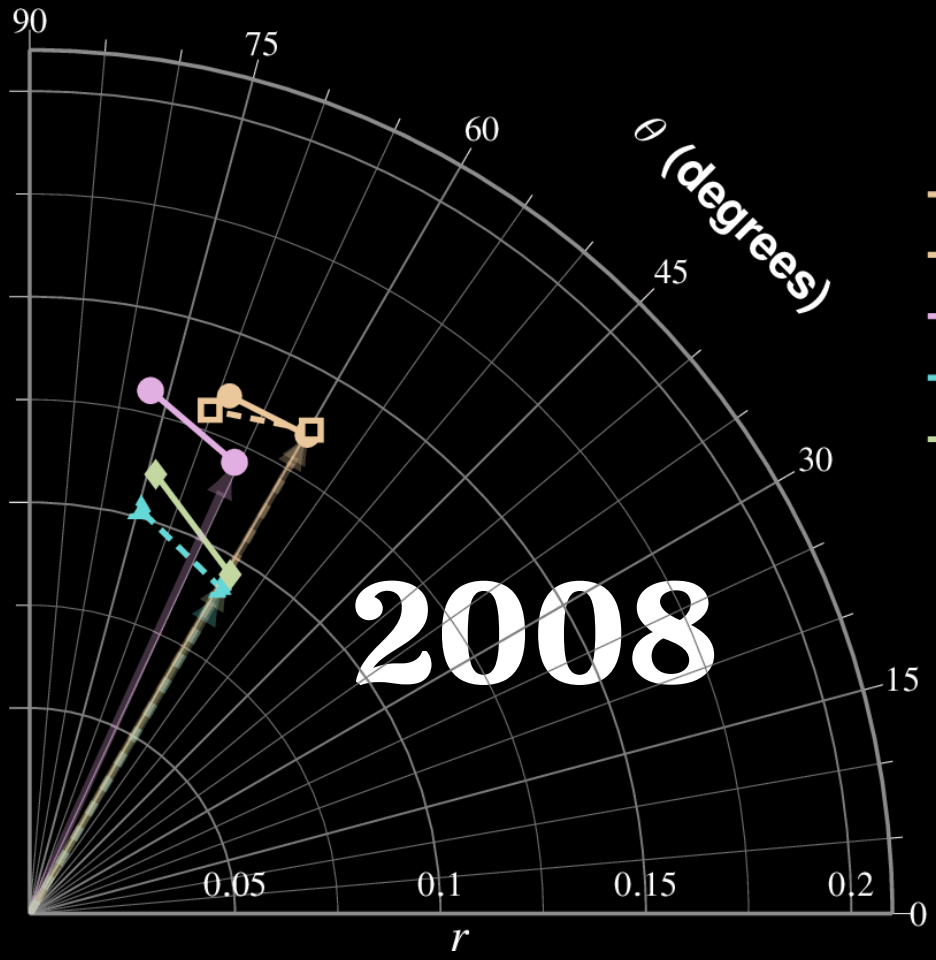
Multiscale spectrum $\rightarrow \mathcal{S}_X = \left\{ \left(\Delta C_X(d), \Delta K_X^{\text{pseudo}}(d) \right) \right\}_d$



The trajectory across depth fingerprints how organization is distributed from persistent to progressively less-persistent structure.

Fingerprinting LSMs

Multiscale spectrum $\rightarrow \mathcal{S}_X = \left\{ \left(\Delta C_X(d), \Delta K_X^{\text{pseudo}}(d) \right) \right\}_d$



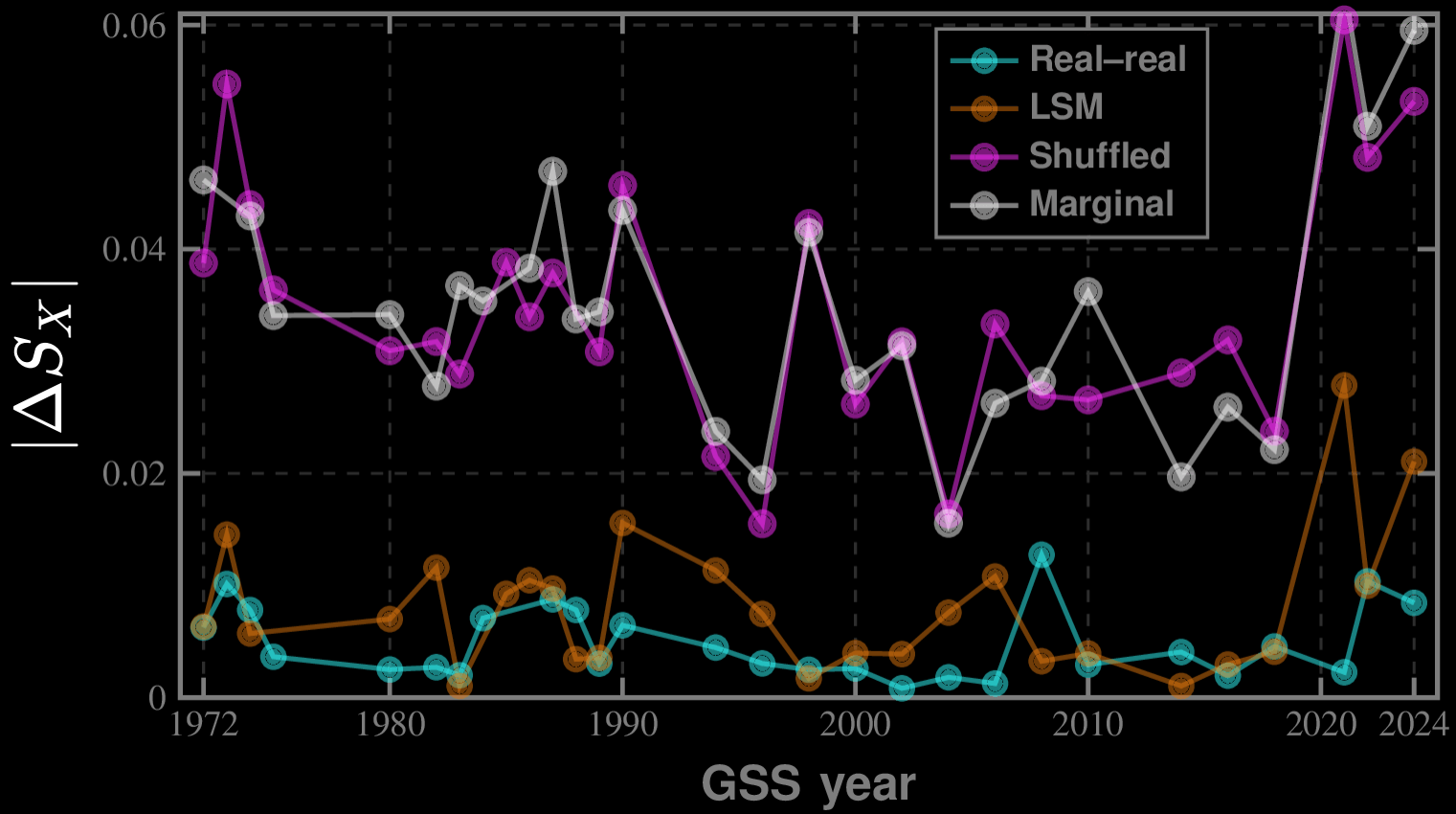
The trajectory across depth fingerprints how organization is distributed from persistent to progressively less-persistent structure.

Validation

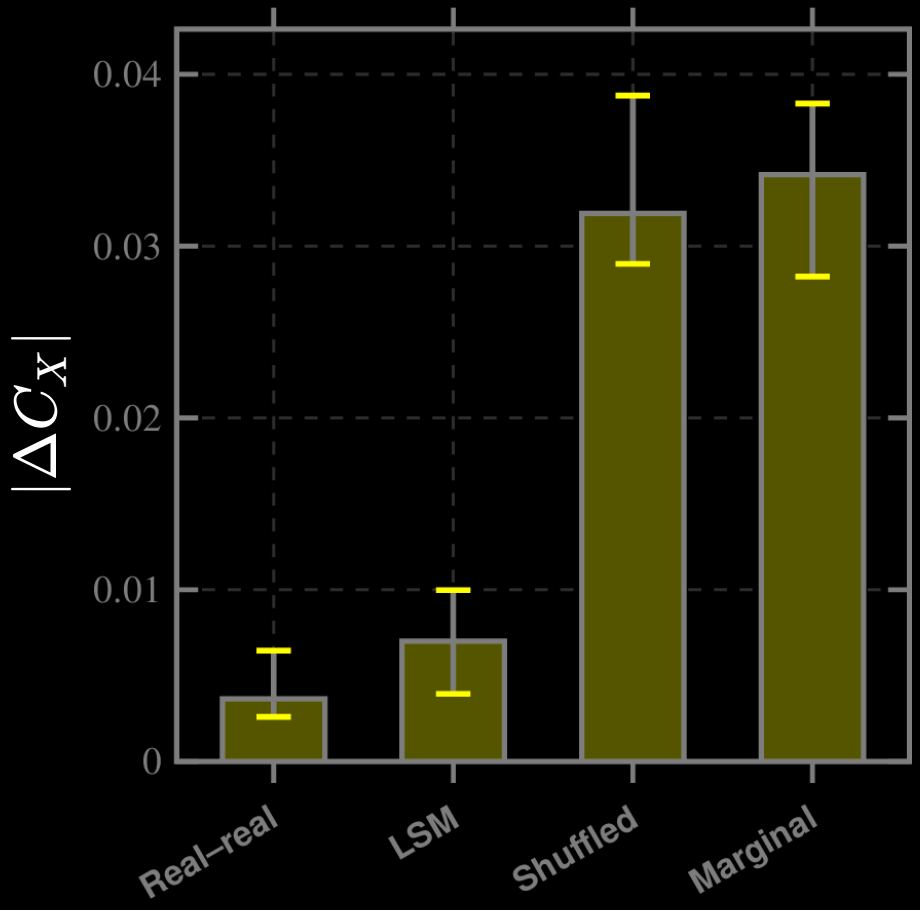
- Multiscale Spectral fingerprinting
- Corruption Recognition
- Synthetic Data Generation

Validating LSMs via Spectral Fingerprinting

Multiscale spectrum $\rightarrow \mathcal{S}_X = \left\{ \left(\Delta C_X(d), \Delta K_X^{\text{pseudo}}(d) \right) \right\}_d$



Validating LSMs via Spectral Fingerprinting



MAP alignment score

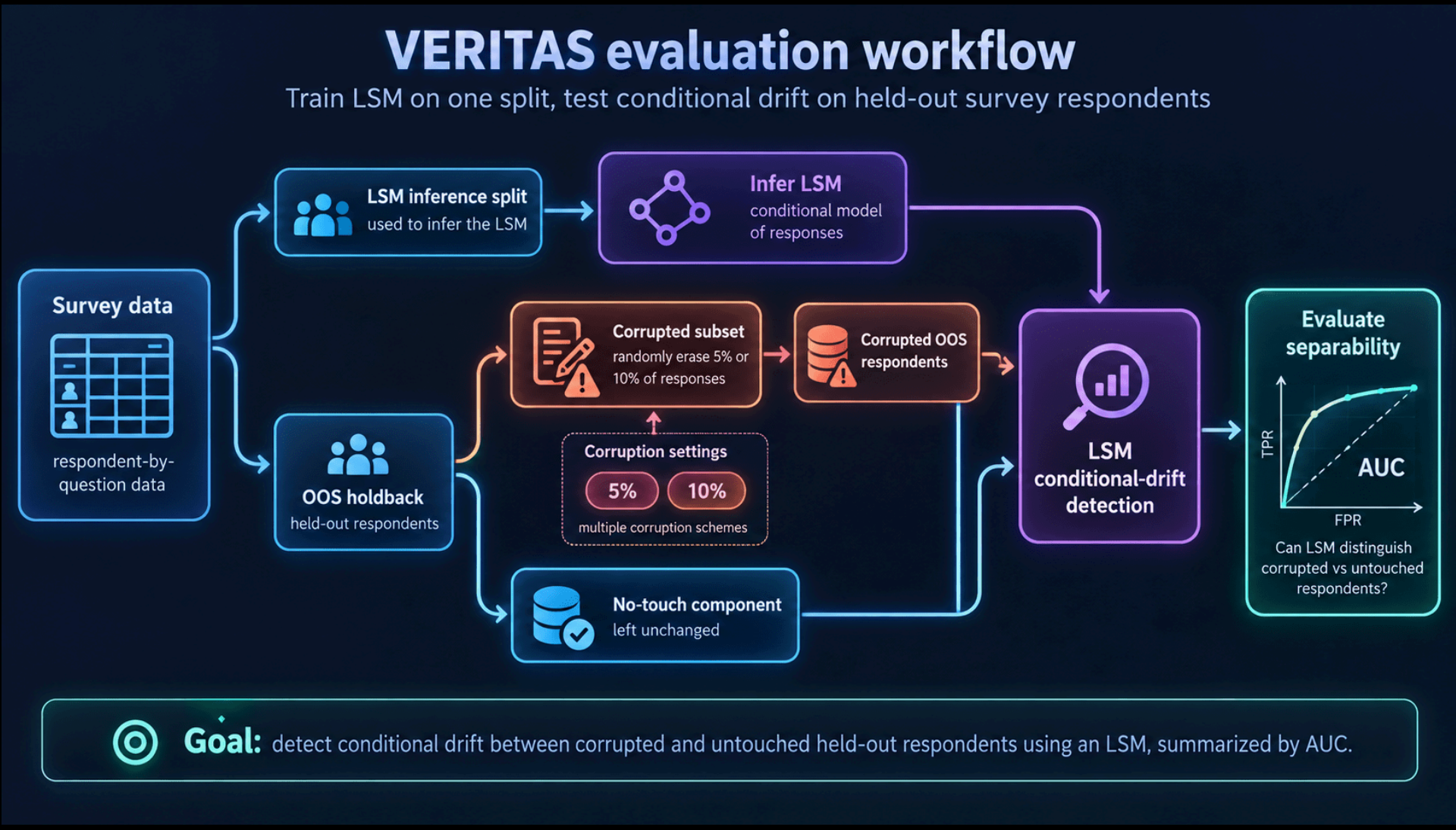
$$u_{X,i}^{(d)} = \frac{\phi_{X,i}^{(d)}(x_i \mid x_{-i})}{\max_{a \in \mathcal{A}_i} \phi_{X,i}^{(d)}(a \mid x_{-i})}$$

$$\Delta C_X(d) = \frac{1}{n} \sum_{i=1}^{n} \left| u_{X,i}^{(d+1)} - u_{X,i}^{(d)} \right|$$

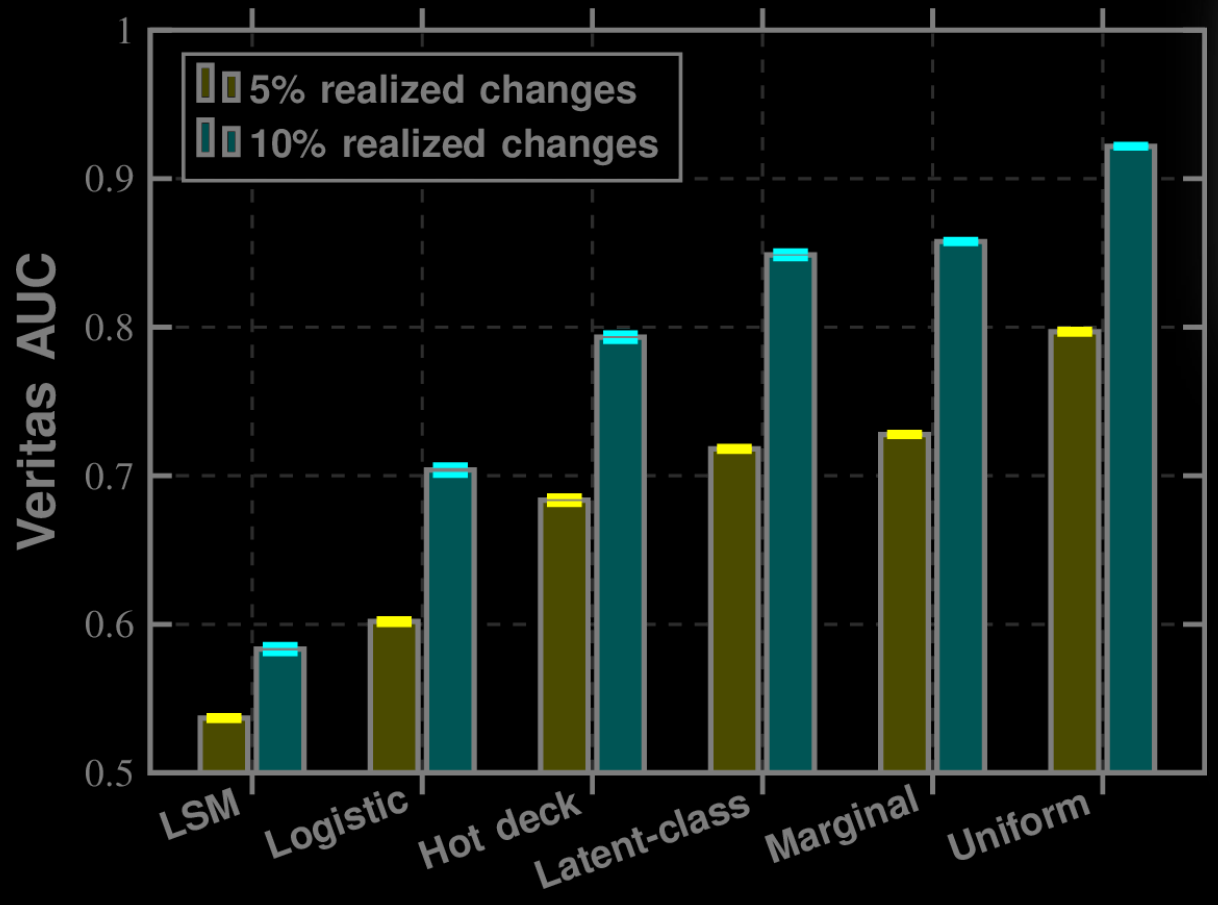
Multi-scale Measure of conditional organization

Validating via Corruption Recognition

VeRITAS: Vetting Response Integrity from cross-Talk in Adversarial Surveys



Validating LSMs Corruption Recognition



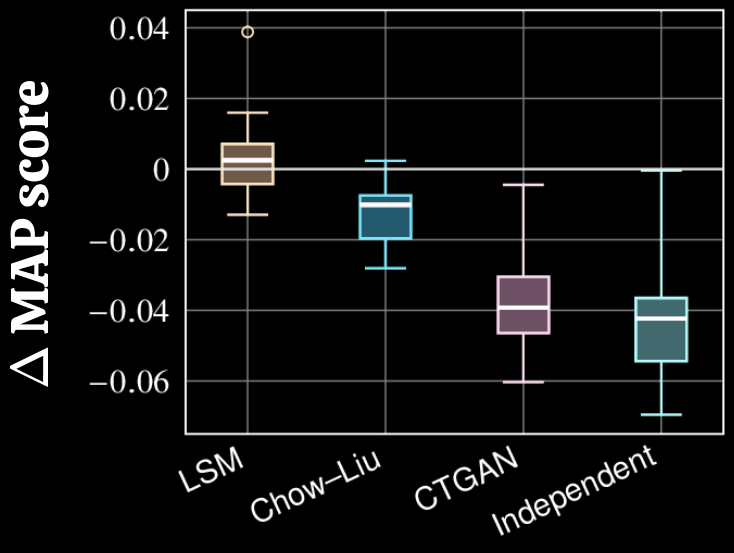
1.

Corruption	AUC
>25%	~100%
~10%	80-90%
~5%	60-80%

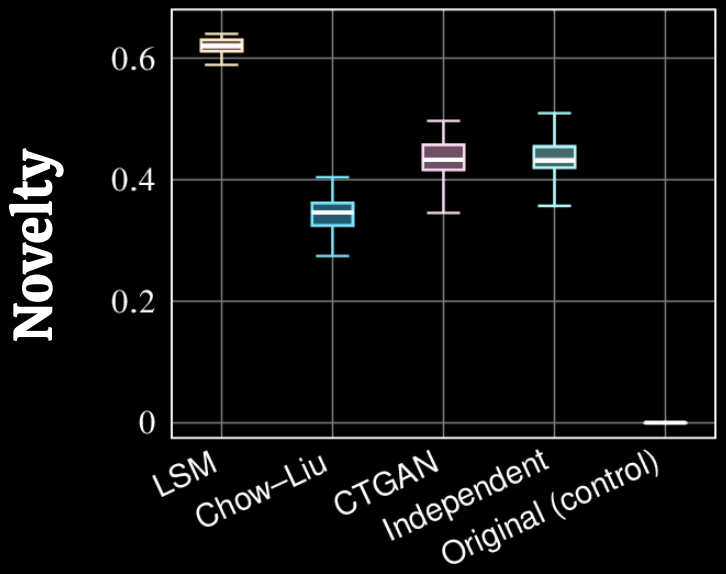
2.

If using LSM to corrupt (LSM-imputation) AUC < 60%

Comparing Against SOTA for Synthetic Data Generation



Conditional Kolmogorov complexity →



Easy to match conditionals if we just shuffle the data! LSM keeps that error low while having high novelty

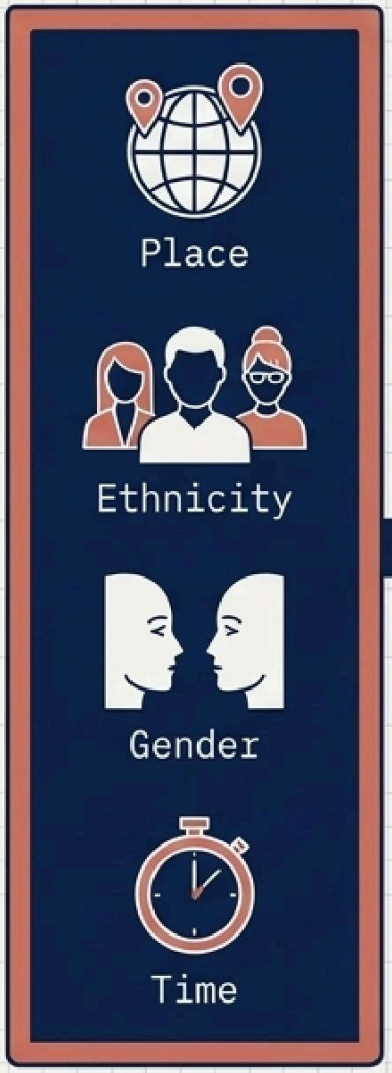
Reflexivity

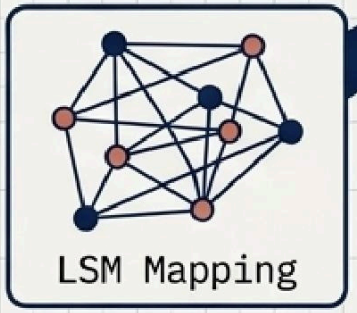
Making observations changes the system

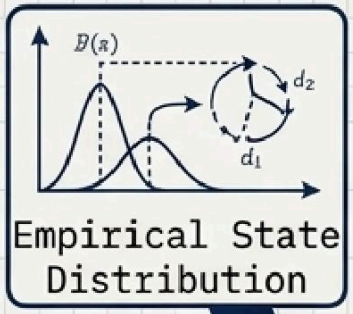
$$x^{-i} \rightarrow y^{-i} \Rightarrow \phi^{i}(x^{i}|x^{-i}) \rightarrow \phi^{i}(x^{i}|y^{-i})$$

https://github.com/zeroknowledgediscovery/psisim

Architecture of DTAG (Digital Twin Anchored Gen

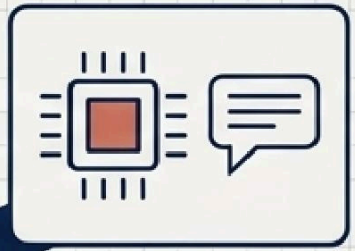






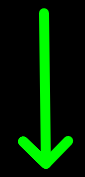
DTAG Cycle





Opinion Control via Reflexivity

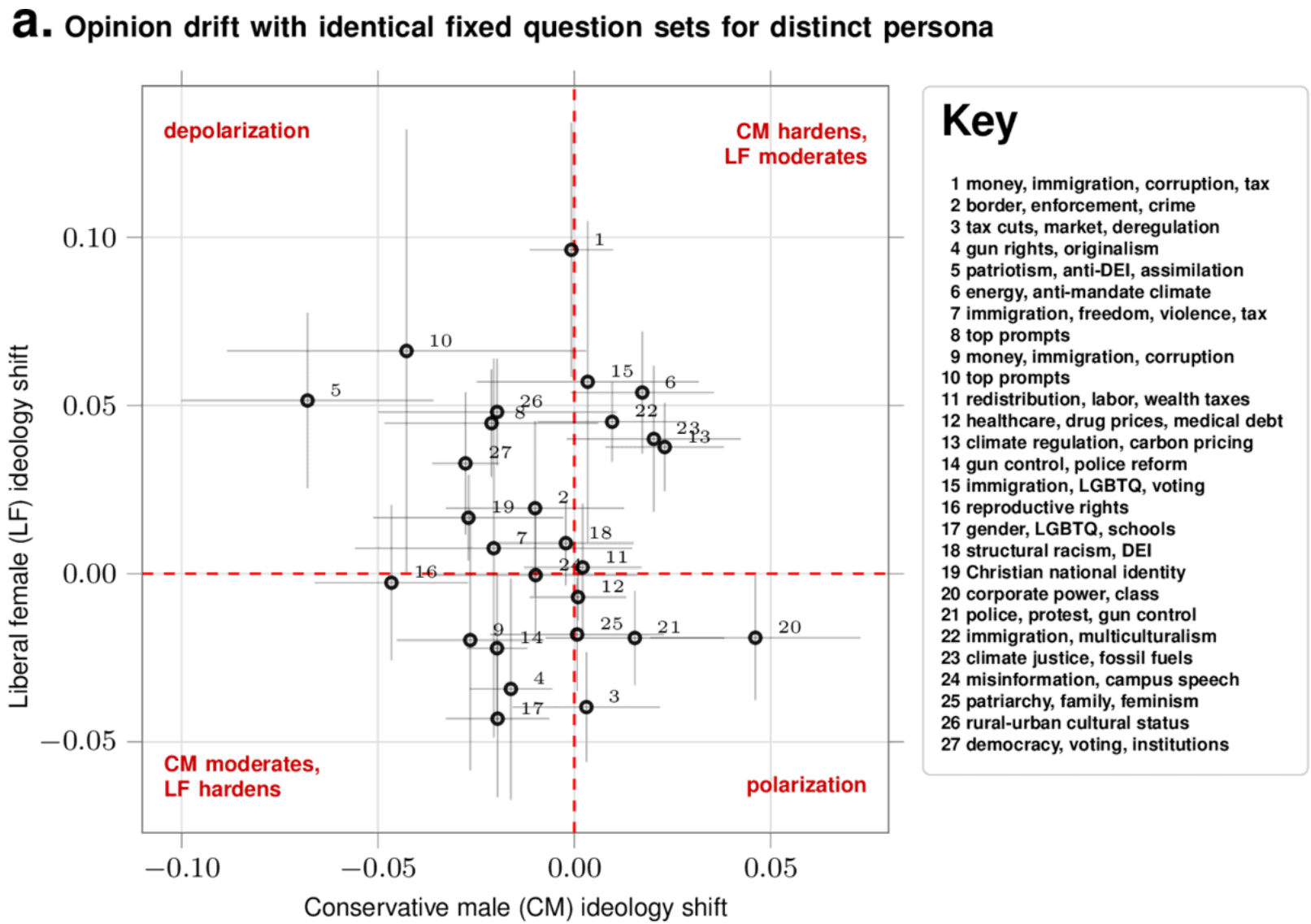
$$x^{-i} \to y^{-i} \Rightarrow \phi^i(x^i|x^{-i}) \to \phi^i(x^i|y^{-i})$$



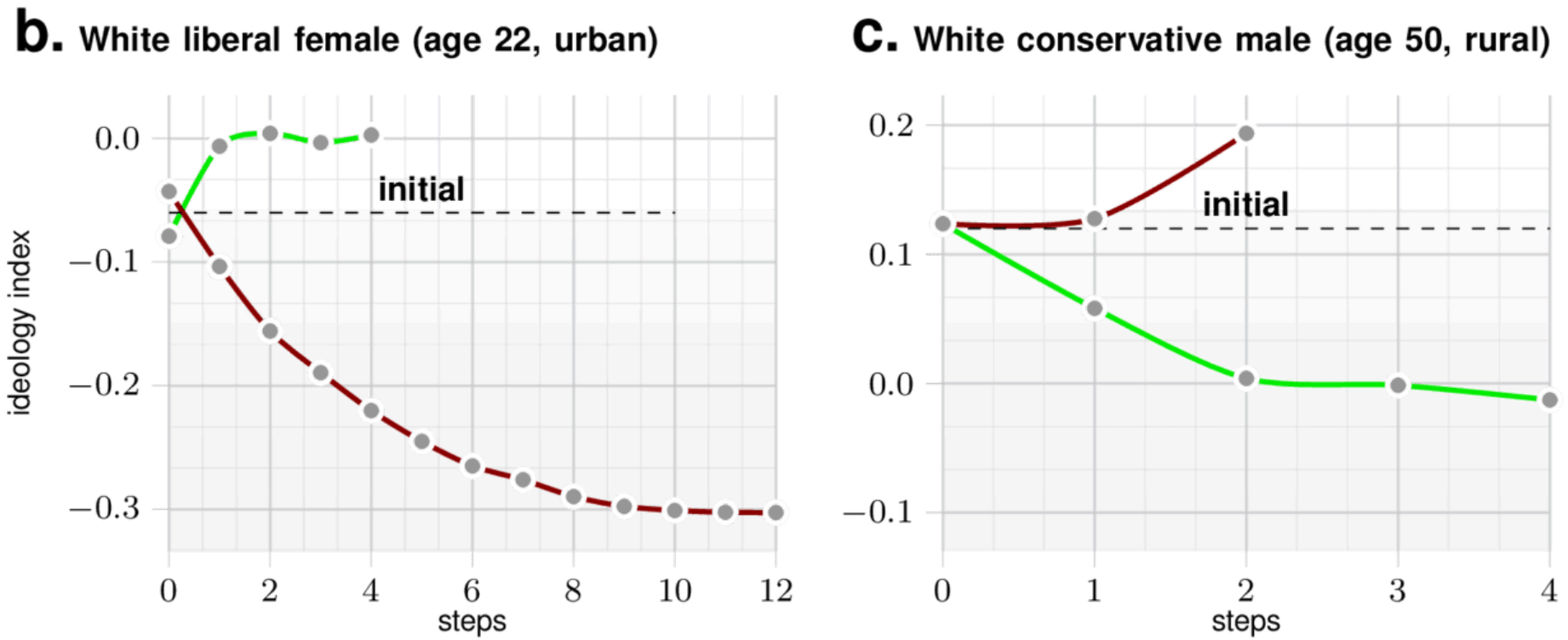
$$y^{-i*} = \arg\max_{y^{-i} \in \mathcal{F}} \left[\phi^i(x^i_\star \mid y^{-i}) - \lambda\, C(x^{-i}, y^{-i})\right]$$

https://github.com/zeroknowledgediscovery/DTAG

- **Fixed Question Sets Exist That Move Different Persona Towards Polarization/Depolarization**



- **We can optimize question sequences to move the same persona in a chosen direction**



- **Note: It is relatively easy to polarize than to depolarize**

Progress & Dissemination

- **Publications:** 1 published (Emergenet), 1 under review, 2 in preparation
- **Conference:** CSS 2026 poster
- **Personnel:** Hari Dahal, PhD — Postdoctoral Scholar

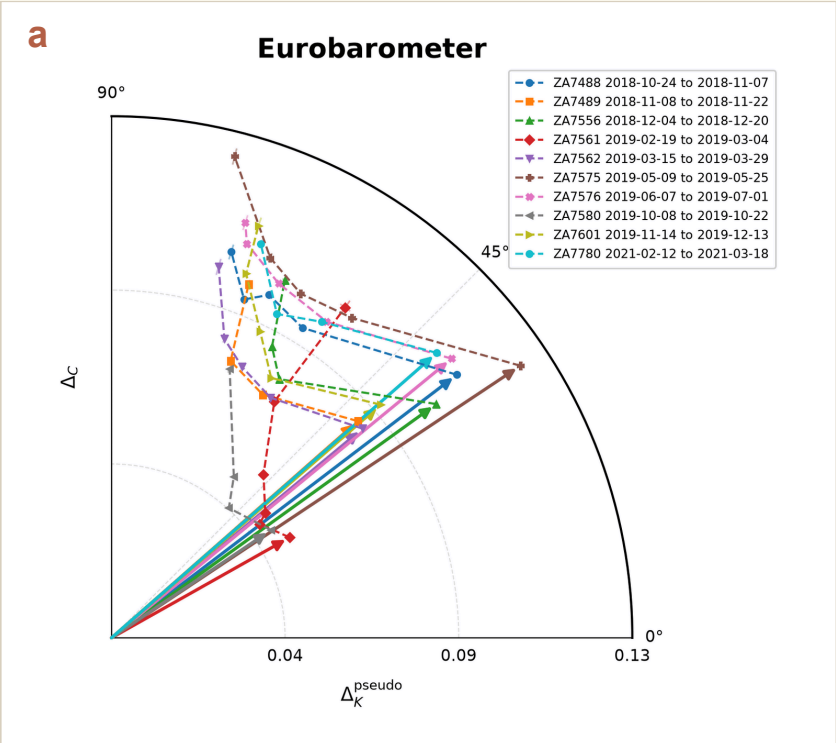
Request: No-Cost Extension

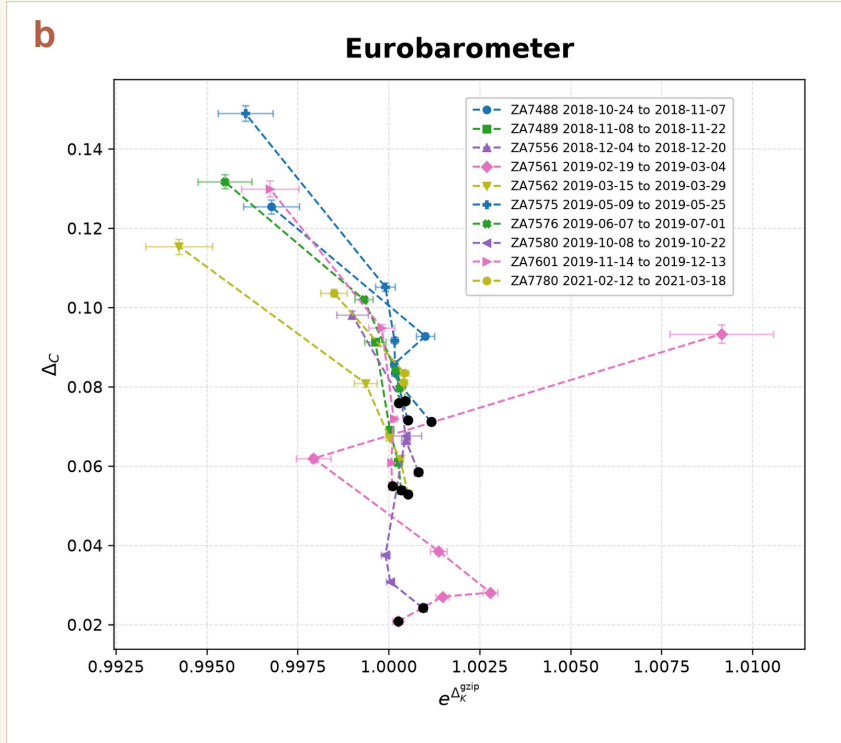
- Postdoctoral start was delayed by ~5–6 months, compressing the planned research period
- If administratively feasible, request a **3–6 month no-cost extension**
- Would allow completion of planned validation, analysis, and dissemination

Potential Follow-on Alignment

- Next steps? Future Programs at IPTO?
- LSMs for trust, influence, and population-level belief dynamics
- Potential alignment with AFOSR **Trust & Influence (Dr. Laura Steckman)**
  https://www.afrl.af.mil/

Eurobarometer: ten waves, 2018–2021





DIRECTION

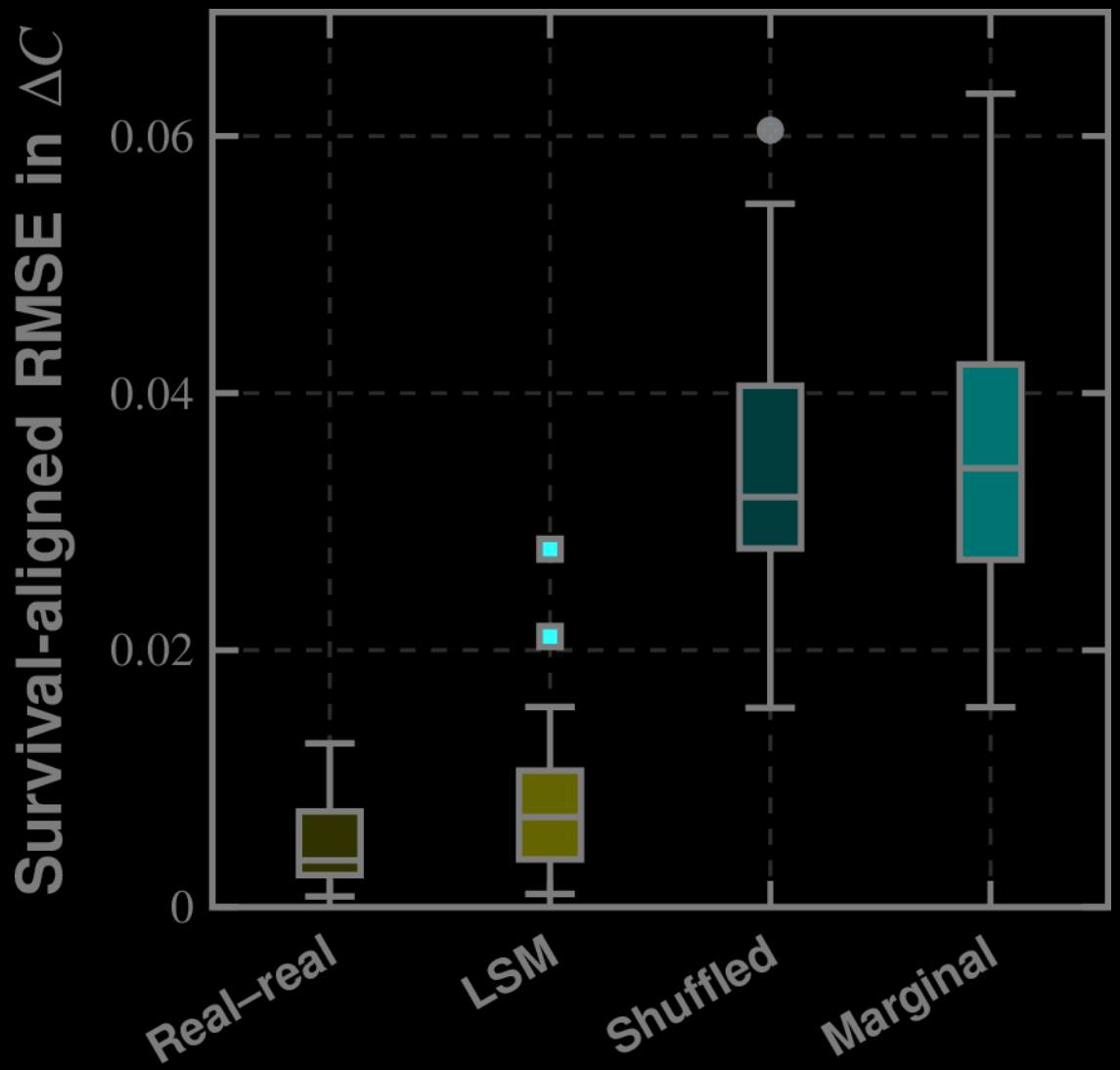
All ten arrows point within ≈30–43° in the (information drift, structural drift) plane.

MAGNITUDE

Arrow lengths range from ≈0.045 to ≈0.12; the 2021 wave, ZA7780, lies within the 2018–19 range.

KOLMOGOROV DRIFT

Within ≈ ±0.006 for every wave except one ZA7561 transition at ≈ +0.009.


Survival-aligned RMSE in ΔC
0.06
0.04
0.02
0
Real–real
LSM
Shuffled
Marginal